

Выгребов Анна

Задача 1.

$$1. \begin{pmatrix} 2 & 1 & 4 & 3 & 5 \\ 5 & 5 & 4 & 8 & 5 \\ 3 & 4 & 5 & 2 & 1 \\ 1 & 5 & 2 & 4 & 3 \\ 5 & 11 & 2 & 11 & 3 \end{pmatrix} = \begin{pmatrix} 0 & -9 & 0 & -5 & -1 \\ 0 & -20 & -6 & -12 & -10 \\ 0 & -11 & -1 & -10 & -8 \\ 1 & 5 & 2 & 4 & 3 \\ 0 & -14 & -8 & -9 & -12 \end{pmatrix} =$$

$$\begin{array}{l} \begin{array}{c} \overline{4} \\ = \end{array} \begin{pmatrix} -9 & 0 & -5 & -1 \\ -20 & -6 & -12 & -10 \\ -11 & -1 & -10 & -8 \\ -14 & -8 & -9 & -12 \end{pmatrix} = -2 \begin{pmatrix} 35 & -3 & 19 \\ 67 & -1 & 30 \\ 94 & -8 & 51 \end{pmatrix} = -2 \begin{pmatrix} -1481 & 71 \\ 67 & -730 \\ -394 & 0 & 10 \end{pmatrix} \end{array}$$

$$= 2 \begin{pmatrix} -148 & -71 \\ -394 & -789 \end{pmatrix} = 4 \begin{pmatrix} -74 & -71 \\ -197 & -789 \end{pmatrix} = 4 \begin{pmatrix} 13986 & -73987 \end{pmatrix}$$

$$= -4$$

$$2. A = \begin{pmatrix} 1 & 2 & 2 & -1 \\ 2 & 3 & 4 & 5 \\ 7 & 3 & 2 & 5 \\ 3 & 2 & 4 & -3 \end{pmatrix} \quad B = \begin{pmatrix} \textcircled{3} & 2 & -1 \\ -1 & \textcircled{5} & 3 \\ -2 & 4 & \textcircled{2} \\ 1 & 3 & 2 \end{pmatrix}$$

k_1
 k_2

$$A|B = \begin{pmatrix} -3 & 2 & 4 \\ -1 & 5 & 22 \\ -2 & -1 & 21 \\ 8 & -21 & -7 \end{pmatrix}$$

$$A = \begin{pmatrix} 4 & 2 & -1 \\ 5 & 3 & -2 \\ 3 & 2 & -1 \end{pmatrix} \quad A^{-1} = \frac{A^5}{\Delta A}$$

$$A_{41} = \begin{vmatrix} 3 & -2 \\ 2 & -1 \end{vmatrix} = 1$$

$$A_{12} = \begin{vmatrix} 5 & -2 \\ 3 & -1 \end{vmatrix} = -1$$

$$A_{13} = \begin{vmatrix} 5 & 3 \\ 3 & 2 \end{vmatrix} = +1$$

$$|A| = 4 \cdot 1 + 2 \cdot (-1) + (-1) \cdot 1 = 1$$

$$A_{21} = \begin{vmatrix} 2 & -1 \\ 2 & -1 \end{vmatrix} = 0$$

$$A_{31} = \begin{vmatrix} 2 & -1 \\ 3 & -2 \end{vmatrix} = -1$$

$$A_{22} = \begin{vmatrix} 4 & -1 \\ 3 & -1 \end{vmatrix} = -1$$

$$A_{32} = \begin{vmatrix} 4 & -1 \\ 5 & -2 \end{vmatrix} = 3$$

$$A_{23} = \begin{vmatrix} 4 & 2 \\ 3 & 2 \end{vmatrix} = -2$$

$$A_{33} = \begin{vmatrix} 4 & 2 \\ 5 & 3 \end{vmatrix} = 2$$

$$A^S = \begin{pmatrix} 1 & 0 & -1 \\ -1 & -1 & 3 \\ 1 & -2 & 2 \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{7} \begin{pmatrix} 1 & 0 & -1 \\ -1 & -1 & 3 \\ 1 & -2 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 2 & -1 \\ 5 & 3 & -2 \\ 3 & 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & -1 \\ -1 & -1 & 3 \\ 1 & -2 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{cases} x_1 + 2x_2 + 3x_3 - 2x_4 + x_5 = 4 \\ 3x_1 + 6x_2 + 5x_3 - 4x_4 + 3x_5 = 5 \\ x_1 + 2x_2 + 7x_3 - 4x_4 + x_5 = 11 \\ 2x_1 + 4x_2 + 2x_3 - 3x_4 + 3x_5 = 6 \end{cases}$$

$$\left(\begin{array}{ccccc|c} 1 & 2 & 3 & -2 & 1 & 4 \\ 3 & 6 & 5 & -4 & 3 & 5 \\ 1 & 2 & 7 & -4 & 1 & 11 \\ 2 & 4 & 2 & -3 & 3 & 6 \end{array} \right) \Rightarrow \left(\begin{array}{ccccc|c} 1 & 2 & 3 & -2 & 1 & 4 \\ 0 & 0 & -4 & 2 & 0 & -7 \\ 0 & 0 & 4 & -2 & 0 & 7 \\ 0 & 0 & -4 & 1 & 1 & -2 \end{array} \right)$$

$$\left(\begin{array}{ccccc|c} 1 & 2 & 3 & -2 & 1 & 4 \\ 0 & 0 & -4 & 2 & 0 & -7 \\ 0 & 0 & -4 & 1 & 1 & -2 \end{array} \right) \Rightarrow \left(\begin{array}{ccccc|c} 1 & 2 & 3 & -2 & 1 & 4 \\ 0 & 0 & -4 & 2 & 0 & -7 \\ 0 & 0 & 0 & -1 & 1 & 5 \end{array} \right)$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 & -1 & 0 & -4 & | & 1 & 2 & 3 & -1 & 0 & | & -1 \\ 0 & 0 & -4 & 2 & 0 & -7 & | & 0 & 0 & 1 & -\frac{1}{2} & 0 & | & -\frac{7}{4} \\ 0 & 0 & 0 & -1 & 1 & 5 & | & 0 & 0 & 0 & -1 & 1 & | & 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 0 & \frac{1}{2} & 0 & -\frac{35}{4} & | & \frac{1}{4} & \frac{25}{4} \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{7}{4} & | & \frac{7}{4} \\ 0 & 0 & 0 & -1 & 1 & 5 & | & 5 \end{bmatrix} \Rightarrow \begin{aligned} x_1 &= -2x_2 - \frac{1}{2}x_4 + \frac{25}{4} \\ x_3 &= \frac{1}{2}x_4 + \frac{7}{4} \\ x_5 &= x_4 + 5 \end{aligned}$$

$$\begin{aligned} 2x_2 - \frac{1}{2}x_4 - \frac{25}{4} + 2x_2 + \frac{3}{2}x_4 + \frac{21}{4} - 2x_4 + x_4 + 5 &= \\ = -\frac{25}{4} + \frac{27}{4} + 5 &= 4 \end{aligned}$$

$$1. \quad x = \begin{bmatrix} 3 \\ 3 \\ -1 \end{bmatrix}, \quad p = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad q = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad r = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 3x_1 \\ x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} -x_2 \\ 2x_2 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2-x_3 \\ 0 \\ 2x_3 \end{bmatrix} \Rightarrow$$

$$\Rightarrow \begin{bmatrix} 3x_1 - x_2 - x_3 \\ x_1 + 2x_2 \\ x_2 + 2x_3 \end{bmatrix} \Rightarrow$$

$$\Rightarrow \begin{cases} 3x_1 - x_2 - x_3 = 3 \\ x_1 + 2x_2 = 3 \\ x_2 + 2x_3 = -1 \end{cases}$$

$$\begin{bmatrix} 3 & -1 & -1 & | & 3 \\ 1 & 2 & 0 & | & 3 \\ 0 & 1 & 2 & | & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 0 & | & 3 \\ 3 & -1 & -1 & | & 3 \\ 0 & 1 & 2 & | & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 0 & | & 3 \\ 0 & -7 & -1 & | & -6 \\ 0 & 1 & 2 & | & -1 \end{bmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 2 & -1 \\ 0 & -7 & -1 & -6 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 13 & -13 \end{pmatrix}$$

$$13x_3 = -13; x_3 = -1$$

$$x_2 + 2 = -1; x_2 = -3$$

$$x_1 + 2 = 3; x_1 = 1$$

$$2. a = \{5, -2, -3, -2\}, b = \{1, 0, 5\}, c_1 = 3a + 9b, c_2 = -a - 3b$$

$$c_1 = (-6, -9, -6) + (9, 0, 45) = (3, -9, 39)$$

$$c_2 = (2, 3, 2) + (-3, 0, -15) = (-1, 3, -13)$$

$$-\frac{3}{-1} = -\frac{9}{3} = -\frac{39}{-13}; \text{ векторы коллинеарные}$$

$$3. a = p - 4q, b = 3p + q; |p| = 1, |q| = 2, (p, q) = \frac{\sqrt{11}}{6}$$

$$\begin{aligned} S &= |[a, b]| = |[p - 4q, 3p + q]| = |[p, 3p] + [4q, 3p] + \\ &+ [p, q] + [4q, q]| = 3[p, p] - 12[q, p] + [p, q] - 4[q, q] = \\ &= |-12[q, p] + [p, q]| = |-11[p, q]| = -11 \cdot 1 \cdot 2 \cdot \frac{1}{2} = \\ &= -11 \end{aligned}$$

$$7. AB = B - A =$$

$$A(3, 2, -4), B(2, -4, -1), C(-6, -5, -8), D(6, 9, 8)$$

$$1. \overline{AB} = B - A = (-1, -6, 3)$$

$$|AB| = \sqrt{1 + 36 + 9} = \sqrt{46}$$

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$$A(6, 2) B(-3, 4) C(14, -7)$$

$$1. AB = \overline{(-9, 2)} \Rightarrow 2x + 9y - 30 = 0$$

$$BC = \overline{(7, -11)} \Rightarrow 11x + 7y + 5 = 0$$

$$CA = \overline{(2, 9)} \Rightarrow 9x - 2y - 50 = 0$$

$$2. \{(x, y) \in \mathbb{R}^2 \mid \begin{cases} 2x + 9y - 30 < 0 \\ 11x + 7y + 5 > 0 \\ 9x - 2y - 50 < 0 \end{cases}\}$$

$$3. h_{A \perp BC}$$

$$y - 2 = \frac{7}{11}(x - 6) \Rightarrow 11(y - 2) - 7(x - 6) = 0 \\ \Rightarrow 11y - 22 = 7x - 42 \Rightarrow 7x - 11y - 20 = 0$$

$$4. M_{AC} = \left(\frac{6+14}{2}, \frac{2+(-7)}{2} \right) = (5, -2.5) \quad B(-3, 4)$$

$$\overline{(-4, 16, 5)} = \overline{(16, -13)}$$

$$16(y - 4) = -13(x + 3) \Rightarrow 16y - 64 = -13x - 39 \Rightarrow 13x + 16y - 25 = 0$$

$$5. \frac{AD \cdot AC}{DB \cdot BC} = \frac{AC}{BC} = \frac{\sqrt{85}}{\sqrt{170}} = \frac{1}{\sqrt{2}} \quad u = \frac{2}{\sqrt{85}}, \frac{9}{\sqrt{85}}$$

$$v = \frac{-2}{\sqrt{170}}, \frac{11}{\sqrt{170}} \quad w = 2\sqrt{2} - 7, 9\sqrt{2} + 11$$

$$(9\sqrt{2} + 11)(x - 4) - (2\sqrt{2} - 7)(y + 7) = 0$$

$$6. m = (16, -13) \quad b = (2\sqrt{2} - 7, 9\sqrt{2} + 11)$$

$$16(2\sqrt{2} - 7) - 13(9\sqrt{2} + 11) = -85\sqrt{2} - 255$$

$$\|m\| = \sqrt{16^2 + (-13)^2} = \sqrt{256 + 169} = \sqrt{425} \approx 20,615$$

$$\|b\| = \sqrt{(2\sqrt{2} - 7)^2 + (9\sqrt{2} + 11)^2} = \sqrt{340 + 170\sqrt{8}}$$

$$\approx 24,09$$

$$\sqrt{2} \approx 1,4142$$

$$b \approx (2,828 - 7; 12,726 + 11) = (-4,172; 23,726)$$

$$m = (16; -13)$$

$$16 \cdot (-4,172) + (-13) \cdot 23,726 \approx -375,19$$

$$\cos \theta = \frac{|m \cdot b|}{\|m\| \cdot \|b\|} \approx \frac{1 - 375,191}{20,615 \cdot 24,09} \approx \frac{375,19}{496,6}$$

$$\approx 0,755 \approx 40,9^\circ$$

$$7. AB: 9(x - 1,5) + 2(y - 3) = 0 \Rightarrow 9x - 2y = 7,5$$

$$AC: 2(x - 5) + 9(y + 2,5) = 0 \Rightarrow 2x + 9y = -12,5$$

$$\begin{cases} 9x - 2y = 7,5 \\ 2x + 9y = -12,5 \end{cases} \Rightarrow x = 0,5; y = -1,5$$

$$2x + 9y = -12,5$$

$$- \text{Угел } P: (0,5; -1,5)$$

$$R = \sqrt{(6 - 0,5)^2 + (2 + 1,5)^2} = \sqrt{5,5^2 + 3,5^2} = \sqrt{435} \approx 6,52 \left(\frac{\sqrt{435}}{2} \right)$$

$$8. b = \frac{19 \cdot (-3) - 2 \cdot 4 - 50}{\sqrt{9^2 + (-2)^2}} = \frac{-185}{\sqrt{85}} = -\sqrt{85} \approx -9,22$$

$$9. S = \frac{1}{2} |6 + 11 + (-3) \cdot (-9) + 4 \cdot (-2)| = \frac{1}{2} |66 + 27 - 8| =$$

$$\frac{85}{2} = 42,5$$

$$10. BC: 11x + 7y + 5 = 0$$

$$x = x_0 - A \cdot \frac{Ax_0 + By_0 + C}{A^2 + B^2} \Rightarrow 6 - 11 \cdot \frac{85}{170} = 0,5$$

$$y = y_0 - B \cdot \frac{Ax_0 + By_0 + C}{A^2 + B^2} \Rightarrow 2 - 7 \cdot \frac{85}{170} = -1,5$$

$$Ax_0 + By_0 + C = 11 \cdot 6 + 7 \cdot 2 + 5 = 85, A^2 + B^2 = 170$$

* Проекция: $(0,5; -1,5)$

$$11. 2x + 9y = 30 \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 15 \\ 0 \end{pmatrix}$$

$$12. AB: 2x + 9y - 30 = 0$$

$$20 + 90 + D = 0$$

$$D = -110$$

$$2x + 9y - 110 = 0$$

$$13. BC: 11x + 7y + 5 = 0$$

$$7x - 11y + D = 0$$

$$-70 + 110 + D = 0 \Rightarrow D = -40$$

$$7x - 11y - 40 = 0$$

$$A(3, 2, -4), B(2, 4, -1), C(-6, -5, -8), D(6, 9, 8)$$

$$1. AB(-1, -6, 3) \quad |AB| = \sqrt{1 + 36 + 9} = \sqrt{46} \approx 6,782$$

$$\begin{array}{l}
 2. \text{ ABC: } \begin{array}{l} \text{AX} \quad x-3 \quad y-2 \quad z+4 \\ \text{AB} \quad -1 \quad -6 \quad 3 \\ \text{AC} \quad -9 \quad -7 \quad -4 \end{array} \\
 \end{array}$$

$$\begin{aligned}
 & 45(x-3) - 31(y-2) - 47(z+4) = 45x - 45 - \\
 & -31y + 62 - 47z - 188 = 0 \Rightarrow 45x - 31y - 47z - \\
 & -171 = 0
 \end{aligned}$$

$$\begin{array}{l}
 3. \quad \begin{array}{|c|c|c|} \hline i & j & k \\ \hline -1 & -6 & 3 \\ -9 & -7 & -4 \\ \hline \end{array} = 45i - 31j - 47k \\
 \sqrt{2025 + 961 + 2209} = \sqrt{5195} \approx 72.076
 \end{array}$$

$$\text{S}_{\text{ABC}} = \frac{1}{2} \sqrt{5195}$$

4. AC.

$$\frac{x-3}{-6-3} = \frac{y-2}{-5-2} = \frac{z+4}{-8+4} \Rightarrow \frac{x-3}{-9} = \frac{y-2}{-7} = \frac{z+4}{-4}$$

$$\begin{cases} x = -9t + 3 \\ y = -7t + 2 \\ z = -4t + 4 \end{cases}$$

$$5. \text{ DH. } \frac{x-6}{45} = \frac{y-9}{-31} = \frac{z-8}{-47}$$

$$\begin{cases} x = 45t + 6 \\ y = -31t + 9 \\ z = -47t + 8 \end{cases}$$

6. ACD

$$\begin{array}{ccc|ccc} AX & x-3 & y-2 & z+4 & x-3 & y-2 & z+4 \\ AC & -6-3 & -5-2 & -8+4 & = -9 & -7 & -4 \\ AD & 6-3 & 9-2 & 8+4 & 3 & 7 & 12 \end{array} = 0$$

$$-56(x-3) + 96(y-2) - 42(z+4) = -56x + 168 + 96y - 192 - 42z - 168 = -56x + 96y - 42z - 192 = 0$$

$$\cos \alpha = \frac{(45-37-47)(-56, 96, -42)}{(95, -37, -47) \cdot (-56, 96, -42)}$$

$$\frac{-2520 - 2976 + 1974}{\sqrt{5195} \cdot \sqrt{14116}} = \frac{-3522}{\sqrt{5195} \cdot \sqrt{14116}}$$

$$7. D(-12, -14, -16)$$

$$A(-3, -7, -12)$$

$$\cos \alpha = \frac{(-12, -14, -16) \cdot (-3, -7, -12)}{(-12, -14, -16) \cdot (-3, -7, -12)}$$

$$\frac{36 + 98 + 192}{\sqrt{596} \cdot \sqrt{202}} = \frac{326}{\sqrt{596} \cdot \sqrt{202}}$$

8. AB

$$\begin{array}{ccc|c} AC & \frac{1}{6} & -7 & -4 \\ AD & 13 & 7 & 12 \end{array} = -196$$

$$V_{ABCD} = \left(\frac{-196}{6} \right) = -\frac{98}{3} = \sqrt{5195} \cdot h \cdot \frac{1}{3} \Rightarrow h = \frac{98}{\sqrt{5195}}$$

$$9. \begin{cases} 45x - 31y - 72z - 171 = 0 \\ x = 45t + 6 \\ y = -31t + 9 \\ z = -47t + 8 \end{cases}$$

$$2025t + 270 + 961t - 279 + 2209t - 376 - 1771 = 5195t - 556 = 0$$

$$t = \frac{556}{5195}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{45 \cdot 556}{5195} + 6 \\ \frac{-31 \cdot 556}{5195} + 9 \\ \frac{-47 \cdot 556}{5195} + 8 \end{pmatrix}$$

10.

$$A_1(-2; -4,5; -4,5)$$

$$AA_1\left(\frac{x-3}{-5} = \frac{y-2}{-6,5} = \frac{z+4}{-0,5}\right)$$

$$B_1(-1,5; -1,5; -6)$$

$$BB_1\left(\frac{x-2}{-3,5} = \frac{y+4}{-2,5} = \frac{z+1}{-5}\right)$$

$$C_1(2,5; -1; -2,5)$$

$$CC_1\left(\frac{x+6}{8,5} = \frac{y+5}{4} = \frac{z+8}{-5,5}\right)$$

$$x = -5t + 3 \rightarrow -5t + 3 = -3, 5t + 2$$

$$x = -3, 5t + 2 \quad -1, 5t = -1$$

$$t = 1, 5 = \frac{3}{2}$$

$$x = -5t + 3 = -\frac{15}{2} + 3 = -\frac{9}{2}$$

$$y = 4t - 5 = 6 - 5 = 1$$

$$z = -5t - 1 = -\frac{15}{2} - 1 = -\frac{17}{2}$$

11. AD: (13, 7, 12) n (45, -31, -72)

$$\sqrt{\cos \angle} = \frac{(45, -31, -72) \cdot (13, 7, 12)}{\sqrt{8170} \cdot \sqrt{362}}$$

$$\sqrt{\cos \angle} =$$

$$\frac{585 - 217 - 864}{\sqrt{8170} \cdot \sqrt{362}} = \frac{-496}{\sqrt{8170} \cdot \sqrt{362}}$$

12. $x = 45t + 6$

$t = 1, H = (51, -22, -39)$

$y = -31t + 9$

$z = -42t + 8$

$$DX \begin{vmatrix} x-6 & y-9 & z-8 \end{vmatrix}$$

$$DH \begin{vmatrix} 45 & -31 & -42 \end{vmatrix} = 0$$

$$DM \begin{vmatrix} 4 & 1 & 2 \end{vmatrix}$$

$$-75(x-6) - 98(y-9) + 169(z-8) = 15x + 90 -$$

$$r = \sqrt{x^2 + y^2} \quad \varphi = \arctg\left(\frac{y}{x}\right); \quad x=2 \quad y=2\sqrt{3}$$

$$r = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{16} = 4$$

$$\varphi = \arctg\left(\frac{2}{2\sqrt{3}}\right) = \frac{\pi}{3}$$

$$x = 4\left(\cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right)\right)$$

$$B. \quad z^3 + 1 = 0$$

$$z^3 = -1$$

$$-1 = 4\left(\cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right)\right)$$

$$z = \sqrt[3]{-1} = \sqrt[3]{1} \left(\cos\left(\frac{\frac{4\pi}{3} + 2\pi k}{3}\right) + i \sin\left(\frac{\frac{4\pi}{3} + 2\pi k}{3}\right) \right)$$

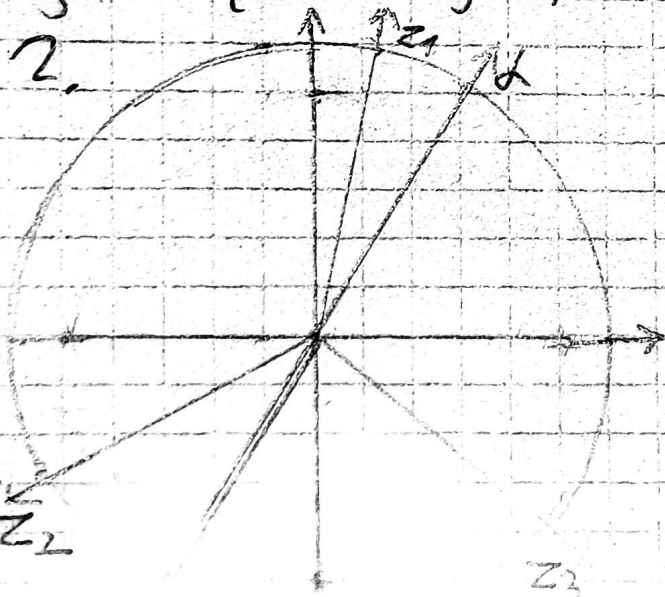
$$+ i \sin\left(\frac{\frac{4\pi}{3} + 2\pi k}{3}\right)$$

$$k=0, 1, 2$$

$$z_1 = \sqrt[3]{1} \left(\cos\left(\frac{4\pi}{9}\right) + i \sin\left(\frac{4\pi}{9}\right) \right)$$

$$z_2 = \sqrt[3]{1} \left(\cos\left(\frac{10\pi}{9}\right) + i \sin\left(\frac{10\pi}{9}\right) \right)$$

$$z_3 = \sqrt[3]{1} \left(\cos\left(\frac{16\pi}{9}\right) + i \sin\left(\frac{16\pi}{9}\right) \right)$$



$$1. \quad 1 \leq |-z+i| \leq 2$$

$$|-z+i|=1$$

$$|-z+i|=2$$

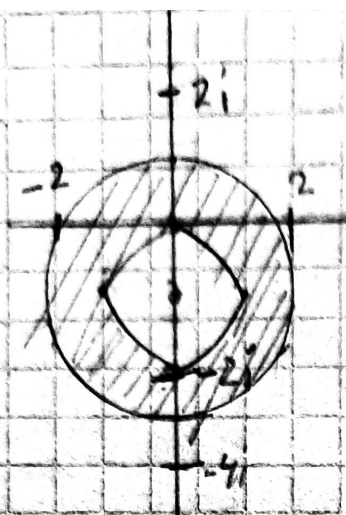
$$|x+iy+i|=|x+i(y+1)|$$

$$\sqrt{x^2+(y+1)^2}$$

$$(y+1)^2$$

$$x^2+(y+1)^2 \leq 2^2$$

$$x^2+(y+1)^2 \geq 1^2$$



$$a = \frac{-1}{-1+i}; \quad b = 243 - 243i$$

$$\frac{-1 - (-1-i)}{(-1+i)(-1-i)} = \frac{1+i-1}{1+1} = \frac{i}{2} = 0,5 + 0,5i$$

$$a+b = 0,5 + 0,5i + 243 + 243i = 243,5 + 243,5i$$

$$a-b = -242,5 + 243,5i$$

$$a \cdot b = (0,5 + 0,5i)(243 - 243i) = 121,5 - 121,5i +$$

$$+ 121,5i - 121,5 = 243$$

$$\frac{a}{b} = \frac{243i}{243^2 + 243^2} = \frac{1}{486} i$$

$$x = \frac{1}{2} \quad y = \frac{1}{2} \quad r = \sqrt{\frac{2}{4}} = \frac{\sqrt{2}}{2}$$

$$\varphi = \arctg(1) = \frac{\pi}{4}$$

$$z = \frac{\sqrt{2}}{2} \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right)$$

$$x = 243 \quad y = -243$$

$$r = \sqrt{118098} = 243\sqrt{2}$$

$$\varphi = \arctan(-1) = -\frac{\pi}{4}$$

$$\beta = 243\sqrt{2}(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4}))$$

$$\angle \beta = \frac{\sqrt{2}}{2} \cdot 243\sqrt{2}(\cos(\frac{\pi}{4} - \frac{\pi}{4}) + i\sin(\frac{\pi}{4} - \frac{\pi}{4})) = 243$$

$$\frac{\angle}{\beta} = \frac{1}{486}$$